Chemistry — Unit 5: Molecular Shape

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Quiz [B] — Bonding Sites, Lone Pairs, Shape Names, and 3-D Drawing

Name: ______________________ Block: ______ Date: ____________

Section 1 — Identify Bonding Sites and Lone Pairs (5.1.1)

For each structural formula shown, count the bonding sites and lone pairs on the central atom. Remember: a double bond or triple bond still counts as ONE bonding site.

1. The structural formula for CO_2 is shown below. Carbon (C) is the central atom. (4 pts)

O = C = O

a. How many **bonding sites** does C have? ______

b. How many **lone pairs** does C have? ______

2. The structural formula for PCl_3 is shown below. Phosphorus (P) is the central atom. (4 pts)

Cl — P — Cl (P also has 1 lone pair; each Cl has lone pairs)

|

Cl

a. How many **bonding sites** does P have? ______

b. How many **lone pairs** does P have? ______

3. Compare NH_3 and NF_3. Both have nitrogen (N) as the central atom. (6 pts)

a. For NH_3: bonding sites = ______ lone pairs = ______ shape = __________

b. For NF_3: bonding sites = ______ lone pairs = ______ shape = __________

c. Do they have the same shape? Explain why or why not:

____

Section 2 — Name Molecular Shapes (5.1.2)

4. Complete the table. Write the shape name and bond angle for each combination. (6 pts — 2 pts per row)

# Bonding Sites	# Lone Pairs	Shape Name	Bond Angle
3	0		
2	1		
3	1		

5. A student says: "H_2S and H_2O have the same molecular shape because both have 2 bonding sites and 2 lone pairs." (6 pts)

Is the student correct? Write your answer below. Include the shape name, bond angles, and explain what lone pairs do to bond angles.

6. A central atom has 4 total electron pairs: 2 bonding sites and 2 lone pairs. (6 pts)

a. What is the **domain geometry** (geometry of all electron pairs including lone pairs)?

____

b. What is the **molecular shape** (based on atom positions only)? ____________________

c. What is the bond angle? ______ °

d. Name one molecule from the workbook that fits this description: ____________________

Section 3 — Draw 3-D Structures Using Wedges and Dashes (5.1.3)

Reminder: ***Regular line*** *= bond in the plane of the paper |* ***Solid wedge (▲)*** *= bond coming toward you |* ***Dashed wedge (- - -)*** *= bond going away from you.*

7. Draw the 3-D structure of CF_4 (carbon tetrafluoride) using wedges and dashes. (8 pts)

Label each bond as "wedge," "dash," or "regular." Place an F at the end of each bond. (4 bonds total)

drawing area

Hint: CF_4 has the same shape as CH_4—the only difference is F instead of H at each end.

8. CHALLENGE — Draw the 3-D structure of H_2O using wedges and dashes. (10 pts)

Include both lone pairs in your drawing (label them or draw two dots for each). Label each bond type.

Then answer: Why is the bond angle of H_2O (104.5°) smaller than the bond angle of CH_4 (109.5°)?

drawing area

Explanation:

Scoring: 3 pts drawing + 3 pts labels + 2 pts lone pairs shown + 2 pts explanation

Total: ______ / 50